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Designing for Impact

Navigating Instructional System Design and Technology Integration Models

Agenda

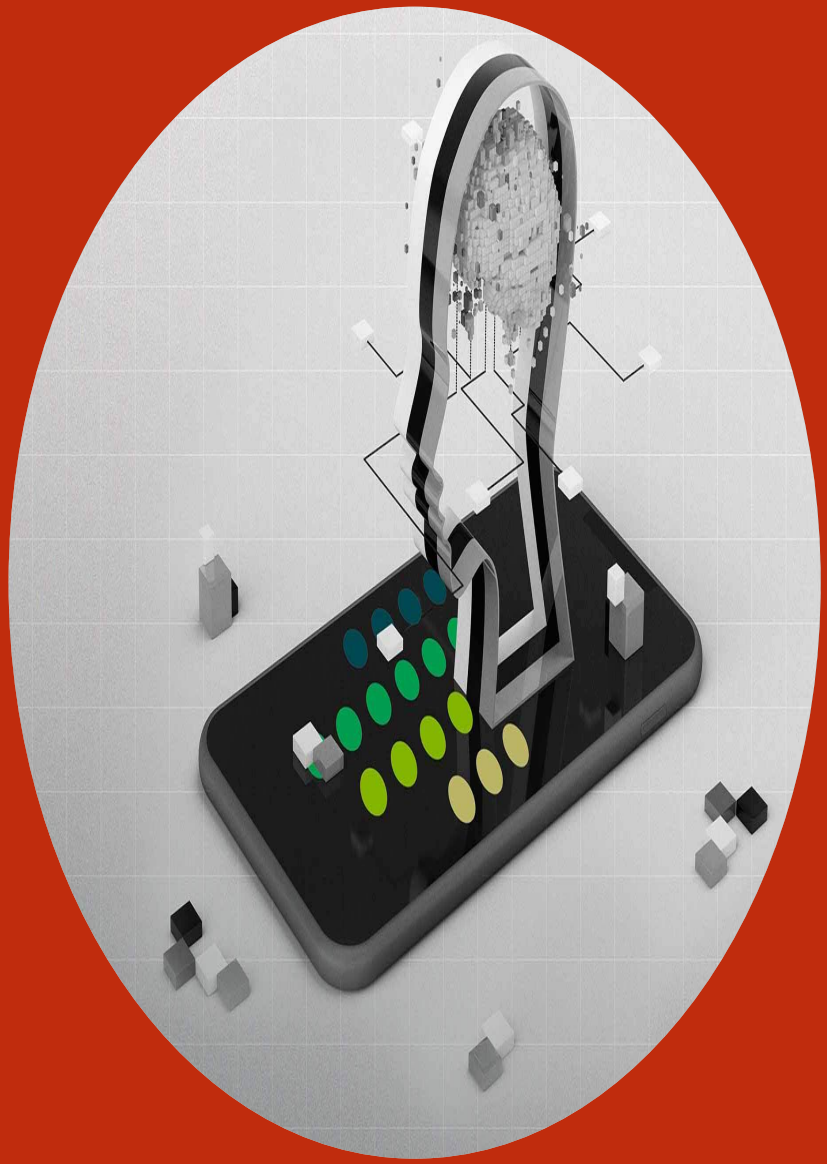
1. ISD Models
   a. ADDIE and PADDIE
   b. ASSURE
   c. Dynamic Instructional Design
2. Technology Integration Models
   a. Florida TIM
   b. SMAR

A teacher buys 30 iPads, lets students “explore” a math app for an hour, but at the end of the day, no once can solve the word problems.

What went wrong?

Was this a ***failure*** of the ***technology*** or the ***instructional design***?

Tech without model is just a distraction.



Instructional Systems Design

What are Instructional Systems Design?

Framework of a Process: systematic, evidence-based process used to create effective education and training programs.

Goal-oriented: involves analyzing learning needs, designing, developing, and evaluating materials to ensure learners achieve specific goals

Core components of ISD

- **Needs Analysis:** Identifying gaps in skills or knowledge and defining learning goals.
- **Design and Development:** Creating learning strategies, content, and materials (e.g., e-learning, workshops).
- **Implementation:** Delivering the training to the target audience.
- **Evaluation:** Measuring the effectiveness of the training and its impact on performance

Why use Instructional Systems Design?

Learner-Centered: Focuses on the needs, prior knowledge, and behavior of the learners.

Consistency and Efficiency: Ensures high-quality training that is cost-effective and saves time.

Systematic Approach: Utilizes structured, step-by-step methods rather than spontaneous design.

We will be discussing **ADDIE** and **PADDIE**, **ASSURE**, and **Dynamic Instructional Design**.

ADDIE & PADDIE

ADDIE & PADDIE

Phases

1. **Analyze:** Identify the learning needs, goals, and target audience.
2. **Design:** Create a blueprint for learning objectives, assessments, and content structure.
3. **Develop:** Build the actual training materials based on the design phase.
4. **Implement:** Deliver the training to the target audience.
5. **Evaluate:** Assess the effectiveness of the training, both during (formative) and after (summative) implementation.

1. **Planning**: Define the project goals, scope, and resource requirements to establish a clear roadmap for the training initiative.
2. **Analysis**: Identify the target audience's current skills, the learning environment, and the specific performance gaps that need to be addressed.
3. **Design**: Map out the learning objectives, assessment methods, and the logical flow of the instructional content.
4. **Development**: Create and assemble the actual learning materials, such as scripts, graphics, and interactive elements, based on the design phase.
5. **Implementation**: Deliver the instructional program to the learners and ensure the necessary facilitators and technologies are prepared.
6. **Evaluation**: Measure the effectiveness of the training by analyzing learner feedback and assessing whether the original performance goals were met.

Why PADDIE?

The "P" for Planning: In PADDIE, the process starts with defining the project's feasibility, budget, timeline, and stakeholders. In ADDIE, these elements are usually buried within the "Analysis" phase or handled outside the model entirely.

Strategic Alignment: PADDIE ensures that the training aligns with organizational goals before a single learning objective is written.

Resource Management: It places a heavier emphasis on identifying who will build the content and what tools are required during the initial "buy-in" stage.

Grade 7 TLE Class (Q1: Information and Communication Technology)

- **Planning (P)**
- **Goal:** Ensure all students can identify hardware and software components.
- **Resources:** Check the availability of the computer lab and ensure the LCD projector is functional.
- **Timeline:** One 60-minute session.
- **Stakeholders:** Coordinate with the ICT coordinator for lab access and technical support.
- **Analysis (A)**
- **Learner Profile:** Grade 7 students with varying levels of prior exposure to computer hardware.
- **Performance Gap:** Students can use devices but often cannot distinguish between system software and application software.
- **Environment:** Computer laboratory setting for hands-on identification.
- **3. Design (D)**

Grade 7 TLE Class (Q1: Information and Communication Technology)

- **3. Design (D)**
- **Learning Competency:** Explain the contributions of different computer system components to overall functionality.
- **Assessment:** A "Hardware Hunt" activity and a 10-item formative quiz.
- **Strategy:** Blended learning using visual aids and actual hardware manipulation.
- **4. Development (D)**
- **Materials:** Create a visual presentation using geometric sans-serif fonts for clarity.
- **Handouts:** Prepare a "Component Checklist" for the laboratory activity.
- **Digital Tools:** Set up a simple interactive drag-and-drop activity on the local network.

Grade 7 TLE Class (Q1: Information and Communication Technology)

- **5. Implementation (I)**
- **Introduction:** Discuss the "Input-Process-Output" (IPO) cycle.
- **Activity:** Students rotate through stations (CPU, RAM, Hard Drive, Monitor) to observe physical parts.
- **Delivery:** Use a projector to show internal components that cannot be safely opened.

- **6. Evaluation (E)**
- **Formative:** Did students correctly identify 80% of the hardware during the "Hardware Hunt"?
- **Reflective:** Collect student feedback on whether the hands-on portion helped them understand abstract concepts like "Processing."
- **Adjustment:** If time was insufficient for the lab tour, plan for a two-day split in the next iteration.

When are they best used?

ADDIE is most effective when the **budget**, **timeline**, and high-level **goals are already set** by an organization or a curriculum board. It is the "gold standard" for:

- **Standardized Education**: Designing school-wide or district-wide curriculum where the objectives are pre-defined by a syllabus (e.g., DepEd or CHED).
- **Small-to-Medium Projects**: When the project scope is simple enough that a separate "Planning" phase would be redundant to the "Analysis" phase.
- **Stable Environments**: When training needs are unlikely to change during the development process.

When are they best used?

PADDIE is effectively used when the teacher is also acting as a **project manager**, say in a Performance Task. It is the preferred choice for:

- **Complex Performance Tasks (PTs):** The **Planning** phase ensures all technical logistics, such as securing the computer lab and verifying the functionality of hardware and software, are finalized before students begin hands-on ICT activities.
- **School-Wide or Grade-Level Assessments:** For high-stakes evaluations like a **National Achievement Test (NAT)** drill, this stage allows for inter-departmental coordination to develop a **Table of Specifications (TOS)** and secure the necessary budget for physical materials.
- **Integration of New Learning Technologies:** When implementing digital learning repositories, the initial phase is used to verify that interactive HTML materials are fully compatible with the specific operating systems and browsers available in the school's laboratory.

ASSURE

ASSURE

Phases

1. **A – Analyze Learners:** Identify the students' general characteristics (age, grade level), specific prior competencies (what they already know), and their preferred learning styles.
2. **S – State Standards and Objectives:** Define exactly what the learners will be able to do after the lesson using the **ABCD** method (Audience, Behavior, Conditions, and Degree).
3. **S – Select Strategies, Technology, Media, and Materials:** Choose the most appropriate instructional methods (e.g., small group work) and the specific digital or physical tools that align with your objectives.
4. **U – Utilize Technology, Media, and Materials:** Follow the "**5 Ps**" process: **P**review the materials, **P**repare the materials, **P**repare the environment, **P**repare the learners, and **P**rovide the learning experience.
5. **R – Require Learner Participation:** Design activities that force students to actively engage with the content, such as hands-on laboratory work, discussions, or interactive digital exercises.
6. **E – Evaluate and Revise:** Assess whether the students met the objectives and reflect on the effectiveness of the media and strategies used to make improvements for the next session.

Grade 7 TLE Class
(Q1: Information and Communication Technology)

- **1. Analyze Learners**
- **General Characteristics:** Grade 7 students (approx. 12–13 years old) with high social digital literacy but low technical hardware knowledge.
- **Prior Competencies:** Students can operate smartphones and basic laptop functions but cannot identify internal components like RAM or the CPU.
- **Learning Styles:** Predominantly visual and tactile learners who benefit from seeing physical hardware.

- **2. State Standards and Objectives**
- **Standard:** Based on the **MATATAG** Curriculum for ICT Quarter 1.
- **Objective:** Given a set of disassembled computer parts (Condition), the Grade 7 students (Audience) will be able to correctly identify and categorize (Behavior) at least 8 out of 10 components (Degree).

Grade 7 TLE Class
(Q1: Information and Communication Technology)

- **3. Select Strategies, Technology, Media, and Materials**
- **Strategies:** Use a "Discovery Learning" approach where students examine parts before the formal lecture.
- **Technology/Media:** A digital slide deck featuring minimalist, geometric typography for high readability.
- **Materials:** Actual system units (opened), RAM sticks, Hard Drives, and a "Component Map" handout.

- **4. Utilize Technology, Media, and Materials**
- **Preview:** Test the interactive HTML-based "Hardware ID" quiz on the lab's Linux workstations.
- **Prepare Environment:** Arrange the computer lab into five "station" zones for small group rotation.
- **Prepare Learners:** Give a "hook" by asking, *"What makes your computer think versus what makes it remember?"*
- **Provide Experience:** Guide students through the hardware stations using the digital presentation as a visual anchor.

Grade 7 TLE Class
(Q1: Information and Communication Technology)

- **5. Require Learner Participation**
- **Activity:** "The Hardware Scavenger Hunt."
- **Engagement:** Students must find the component described on their worksheet within the physical system unit and record its serial number or key features to prove they located it.
- **6. Evaluate and Revise**
- **Assessment:** A quick formative exit ticket using the interactive digital repository.
- **Revision:** If students struggled to differentiate between the Hard Drive and the SSD, plan to bring in a "cut-away" model or a clearer diagram for the next class section.

When is it best used?

- **Technology-Rich Lessons**: Use this when your lesson involves specific digital tools, such as using an interactive HTML-based repository, a simulation, or a specialized software suite for an ICT performance task.
- **Student-Centered Environments**: It is ideal when you want to move away from "chalk and talk" toward active participation, as the "R" (Require Learner Participation) phase forces you to design activities that engage the students directly with the media.
- **Differentiated Instruction**: Because the model starts with "A" (Analyze Learners), it is the best choice when you have a diverse class and need to tailor your media—such as choosing high-contrast, geometric typography for better readability—to fit specific learning styles or needs.
- **Refining Existing Materials:** Use it when you already have a curriculum but want to "ASSURE" that the way you are delivering it—using your school’s laboratory workstations or specific Linux-based tools—is actually effective for your Grade 7 students.

Dynamic Instructional Design

DID

A ***flexible***, ***continuous***, ***non-linear*** approach to lesson planning that emphasizes the evolving relationship between the teacher, the learner, and the technology.

A lesson plan is a "living" document that should be ***constantly adjusted*** based on student performance and feedback.

6 steps:

1. **Know the Learners**: Analyze the students' prior knowledge, cognitive styles, and physical needs to tailor the lesson to their actual "Foundations First" readiness.
2. **State your Objectives**: Define clear, measurable goals using the ABCD (Audience, Behavior, Condition, and Degree) method to ensure mastery of specific skills.
3. **Establish the Learning Environment**: Organize the physical space, such as the ICT lab layout, and the psychological climate to facilitate focused, collaborative learning.
4. **Identify Teaching and Learning Strategies**: Select the most effective pedagogical methods, like direct instruction or hands-on performance tasks, to bridge the gap between objectives and outcomes.
5. **Identify and Select Support Technologies**: Choose specific digital tools and media—such as interactive HTML repositories or geometric sans-serif presentations—that directly support the chosen strategies.
6. **Evaluate and Revise**: Assess both student performance and the instructional design itself, using the data to immediately adjust and improve the next cycle of learning.

Grade 7 TLE Class
(Q1: Information and Communication Technology)

- **1: Know the Learners**
- Identify that Grade 7 students are "digital natives" but lack "foundations first" technical knowledge of internal hardware.
- Note that students prefer visual, hands-on learning over long lectures.
- Recognize that students in the ICT lab have varying typing speeds and basic computer literacy.

- **2: State your Objectives**
- **Audience:** Grade 7 ICT students.
- **Behavior:** Identify and explain the function of internal computer components (CPU, RAM, Storage).
- **Condition:** Given a dismantled system unit and a digital guide.
- **Degree:** With 100% accuracy in physical identification.

Grade 7 TLE Class
(Q1: Information and Communication Technology)

- **3: Establish the Learning Environment**
- **Physical:** Arrange the laboratory workstations in a "U-shape" to allow the teacher to see all screens while leaving the center open for hardware demonstrations.
- **Psychological:** Foster a "Growth Mindset" where students feel safe to ask "why" a component might fail, encouraging troubleshooting and high-order thinking.

- **4: Identify Teaching and Learning Strategies**
- Use **Direct Instruction** to introduce the Input-Process-Output (IPO) cycle.
- Implement **Cooperative Learning** by pairing students for a "Hardware Discovery" task.
- Utilize **Scaffolding** by providing a labeled diagram before asking students to identify parts on a real machine.

Grade 7 TLE Class
(Q1: Information and Communication Technology)

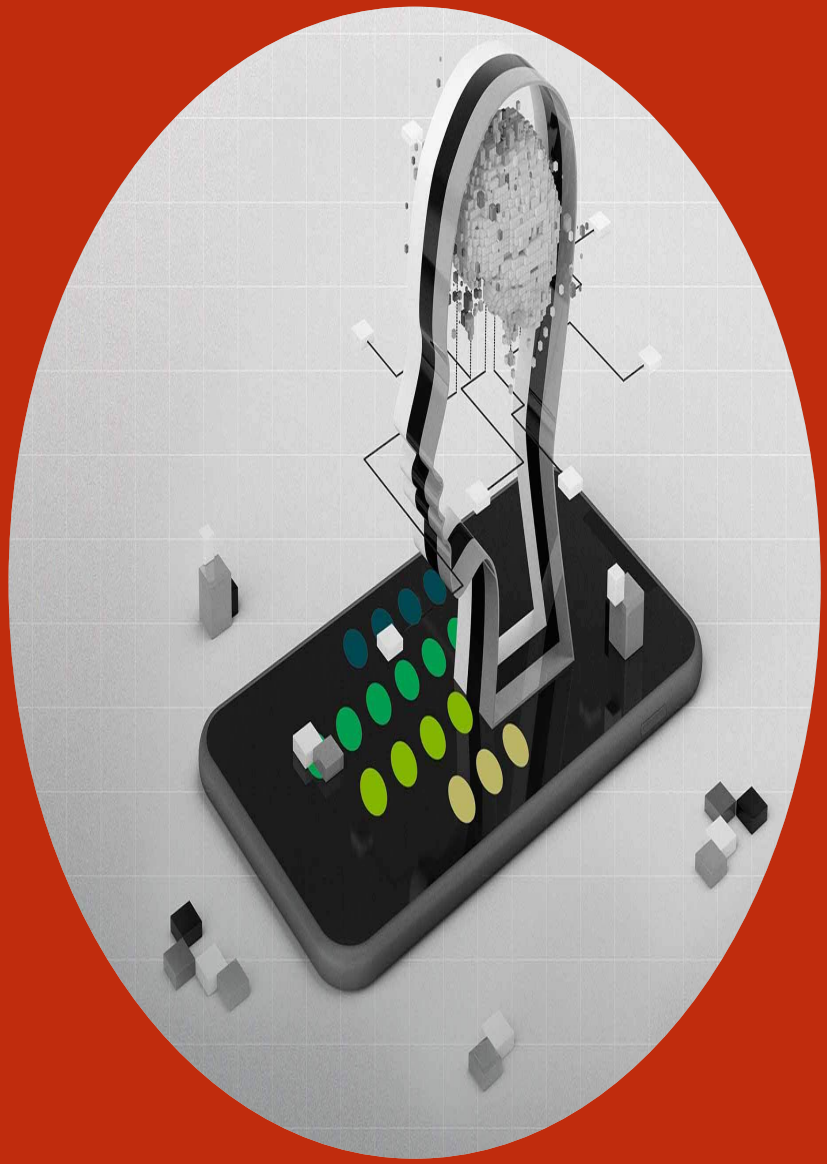
- **5: Identify and Select Support Technologies**
- Deploy an **interactive HTML repository** accessible via the local network for self-paced review.
- Use a digital presentation with **geometric sans-serif typography** to ensure technical terms are easy to read from the back of the lab.
- Provide physical "non-working" system units for students to touch and examine without risk of damaging active equipment.

- **6: Evaluate and Revise**
- **Student Assessment:** Conduct a hands-on "Point and Name" practical exam.
- **Design Evaluation:** If students consistently struggle with the difference between RAM and Storage, revise Step 4 to include a more vivid analogy (e.g., "Desk space vs. Filing cabinet").
- **Technical Check:** Ensure the HTML files ran smoothly on the lab’s Linux-based browsers and adjust code if any lag occurred.

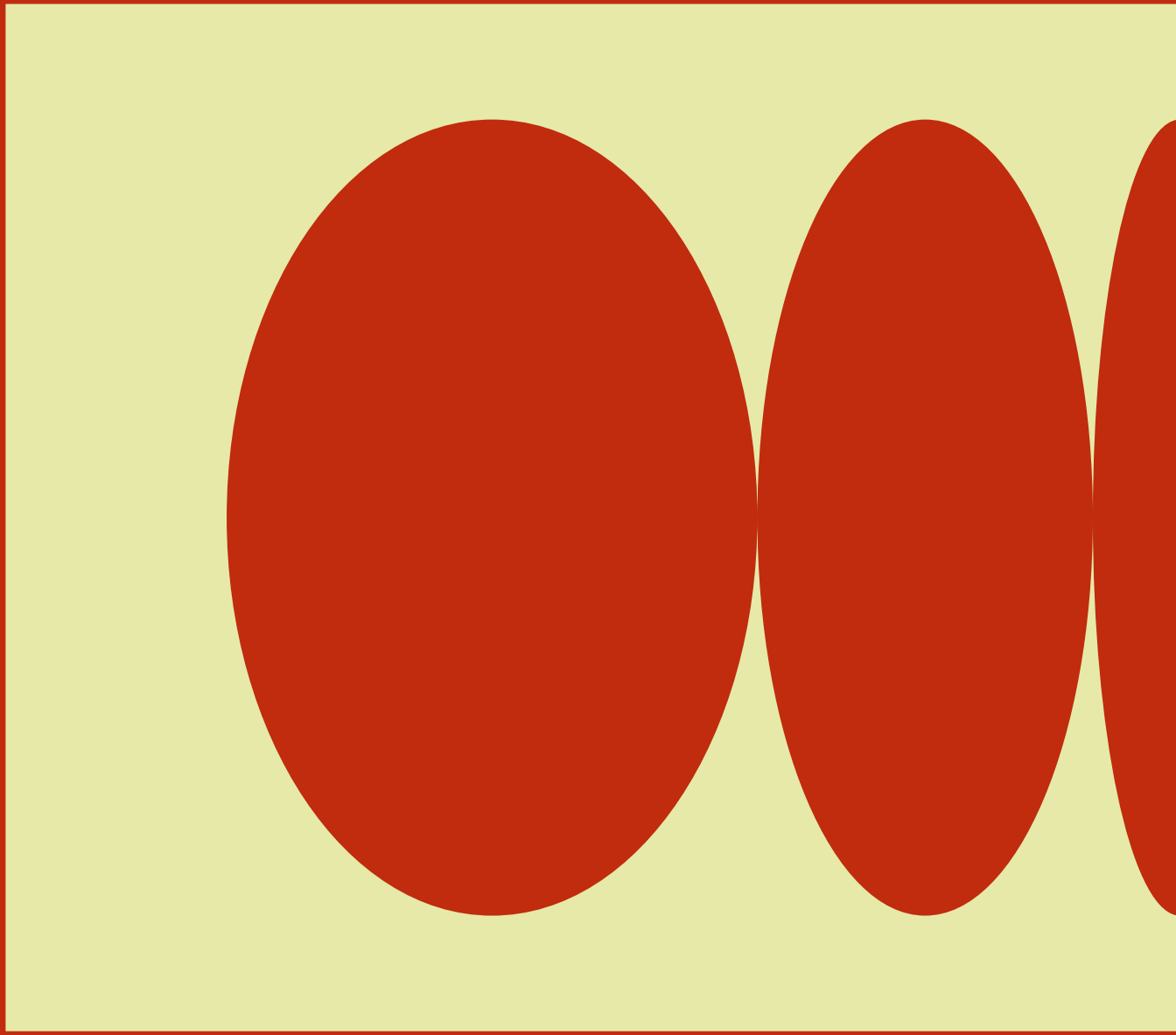
Once we have the blueprint, we have to decide:

Are we just replacing a chalkboard with a screen, or are we fundamentally changing how students learn?

To answer that, we move from the process of design to the **models of integration**.



We will be discussing
SAMR, and
Florida TIM.



SAMR

SAMR

developed by Dr. Ruben Puentedura, is a framework designed to help educators evaluate how they are integrating technology into their teaching.

Substitution (Enhancement)

Technology acts as a direct tool substitute with **no functional change** to the task.=

- **Example:** Instead of students writing an essay with pen and paper, they type it into a word processor.

Augmentation

Technology acts as a direct tool substitute, but with **functional improvements** that make the task more efficient.

- **Example:** Students use the word processor’s built-in spell checker, thesaurus, and "voice-to-text" features to refine their writing.

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Modification

Technology allows for **significant task redesign**, changing how the learning actually happens.

Example: Instead of an individual essay, students use a shared Google Doc to write a collaborative piece, providing real-time peer feedback through the "comments" feature.

Redefinition

Technology allows for the **creation of new tasks** that were previously inconceivable without it.

Example: Students create a multimedia digital documentary or a website that includes interviews conducted via video call with experts from another country, then publish it for a global audience.

Why use SAMR?

- **A tool for reflection:** “Am I just using a computer to do what I used to do on paper, or am I using it to open new doors for my students?”

Florida TIM

Florida Technology Integration Matrix

developed by the Florida Center for Instructional Technology (FCIT)

Unlike the SAMR model (which is a 4-level "ladder"), the TIM is a **5x5 matrix** that intersects 5 levels of integration with 5 characteristics of the learning environment, creating 25 distinct cells for evaluation.

The 5 Levels of Integration

- **Entry:** The teacher uses technology primarily to deliver content to students (e.g., a lecture with a digital slide deck).
- **Adoption:** The teacher directs students in the conventional and procedural use of technology tools (e.g., a guided lab where everyone follows the same steps).
- **Adaptation:** The teacher facilitates students as they begin to use technology tools independently and explore different ways to complete a task.
- **Infusion:** The teacher provides the context, and students choose the appropriate technology tools to achieve the learning outcome.
- **Transformation:** The teacher encourages innovative, unconventional uses of technology to solve problems that would be impossible without it.

Grade 7 TLE Class
(Q1: Information and Communication Technology)

- **1. Entry**
- The teacher uses a projector to **display a digital slide** to deliver a lecture on the "Foundations First" principles of computer hardware.
- **2. Adoption**
- Students **follow a step-by-step guided tutorial** on their Linux workstations to practice creating and naming a specific set of folders in a predetermined directory.
- **3. Adaptation**
- After learning basic file management, students **independently choose how to use a file manager** to organize a provided set of scattered digital assets for a mock school project.
- **4. Infusion**
- Students **select from various digital tools**, such as an interactive HTML repository or a spreadsheet, to monitor their own progress and document the troubleshooting steps they took during a hardware assembly task.

Grade 7 TLE Class
(Q1: Information and Communication Technology)

- **5. Transformation**
- Students use **high-level collaboration tools to co-design an interactive technical support database** that helps peers across the school solve common ICT problems in real-time.

<u>ISD</u>
ADDIE/PADDIE
ASSURE
Dynamic
Instructional Design

<u>TIM</u>
SAMR
Florida TIM

Group Activity

Design your own PeTa!

https://docs.google.com/document/d/10r9CdbWQMrGnslt8MrArF1ZaP2b1JFngoiYcpaaVVSk/edit?usp=sharing

ADDIE or ASSURE?

(Building the System) (Knowing your Audience)

If you are choosing a YouTube video for your class and you're worried that your students will get bored or confused by the narrator's accent, which model are you using to solve that problem

ASSURE

It focuses on the learner's mind (Are they bored? Are they ready?) and the media's impact (Is this video too hard? Is the font too small?) "

3 Key Takeaways

Design Before Tools: ISD models (ADDIE/ASSURE) are the foundation; technology is the aesthetic and functional finish.

SAMR is a Journey: You don’t always have to "Redefine," but you should always strive to move past "Substitution."

Context is King: Use Florida TIM to evaluate *how* students are using tech (e.g., Are they being "Active" or just "Passive"?).

Thank you!

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